Dataset Overview:

Face Mask Dataset:

The dataset is used for a binary image classification task to detect whether a person is wearing a face mask or not. Each image belongs to one of two classes: *with_mask* or *without_mask*.

Training and Validation Dataset

- Used for training and validating the CNN model.
- Images are organized into two folders:
  - with_mask ○ without_mask
- Total images: approximately **7,500** ○ Training images: approximately **6,000** ○ Validation images: approximately **1,500**
- These images are used during model training to learn and monitor performance.

LINK: https://www.kaggle.com/datasets/omkargurav/face-mask-dataset?resource=download

Test Dataset (External)

- Used only for final evaluation of the trained model.
  - Stored separately from training and validation data.
- Total images: **337** ○ with_mask images: present ○ without_mask images: present
- This dataset is not used during training or validation.

LINK: https://www.kaggle.com/datasets/sumansid/facemask-dataset

Optical Character Recognition (OCR):

- The dataset contains scanned handwritten English forms written by different individuals.
- It is used for an OCR task to extract text from handwritten images.
- A small sample of images (around 10–20) is used instead of the full dataset.
- These images are processed and passed to the Tesseract OCR engine to convert them into digital text.

LINK: https://www.kaggle.com/datasets/naderabdelghany/iam-handwritten-forms-dataset?select=__notebook_source__.ipynb

Project - Facial Emotion Recognition:

- The project uses the **FER-2013 dataset**, which contains grayscale images of human faces for emotion classification.
- Each image is of size **48 × 48 pixels**, making it suitable for CNN-based models.
- The dataset consists of around **35,000 images** divided into training and testing sets.
- It includes **7 emotion classes**: Angry, Disgusted, Fearful, Happy, Neutral, Sad, and Surprised.
- Images are organized into **separate folders by emotion**, enabling easy loading using directory-based methods.
- This dataset is widely used for **facial emotion recognition tasks** and provides a good benchmark for model performance.

Testing on Happy Folder (End Part):

- After training the model, predictions were performed on images from the **"happy" test folder**.
- Each image was preprocessed (grayscale, resized to 48×48, and normalized) before being passed to the model.
- The model generated predictions by selecting the emotion with the **highest probability**.

LINK: **https://www.kaggle.com/datasets/ananthu017/emotion-detection-fer**